Pharmacology

Here are 20 multiple-choice nursing questions on Pharmacology, formatted as requested:

1. Which term describes the process by which a drug moves from its site of administration into the bloodstream?
    A. Metabolism
    B. Distribution
    C. Absorption
    D. Excretion
    Answer: C
    Rationale: Absorption is the process by which the drug enters the systemic circulation from its site of administration. Metabolism refers to the breakdown of the drug, distribution is how the drug spreads throughout the body, and excretion is how the drug is eliminated.

2. The primary site for drug metabolism in the body is the:
    A. Kidneys
    B. Lungs
    C. Liver
    D. Brain
    Answer: C
    Rationale: The liver contains enzymes that are crucial for metabolizing (breaking down) drugs, making them easier to excrete. While other organs can metabolize drugs to some extent, the liver is the primary site.

3. Which term refers to the chemical alteration of a drug by the body?
    A. Pharmacodynamics
    B. Pharmacokinetics
    C. Metabolism
    D. Bioavailability
    Answer: C
    Rationale: Metabolism is the process by which the body chemically modifies drugs, often to make them inactive or prepare them for excretion. Pharmacodynamics and pharmacokinetics are broader terms related to drug action and movement, respectively. Bioavailability refers to the fraction of an administered dose of unchanged drug that reaches the systemic circulation.

4. A drug that binds to a receptor and produces a response is called a(n):
    A. Antagonist
    B. Agonist
    C. Inhibitor
    D. Synergist
    Answer: B
    Rationale: An agonist is a substance that binds to a receptor and elicits a biological response, mimicking the action of the endogenous substance. An antagonist blocks the action of agonists. Inhibitors block enzyme activity, and synergists enhance the action of other drugs.

5. Which of the following is an example of an "idiosyncratic reaction" to a drug?
    A. Nausea after taking an antibiotic
    B. An expected allergic rash developing after a penicillin injection
    C. A patient experiencing severe hallucinations after taking a benzodiazepine, when typically it causes sedation

D. Increased blood pressure after taking a decongestant
Answer: C
Rationale: An idiosyncratic reaction is an unusual, unpredictable adverse drug reaction that differs from a typical response and is not an allergic reaction or overdose. The hallucinations in this case are an unexpected and atypical response to the benzodiazepine.

6. The "half-life" of a drug refers to:
A. The time it takes for the drug to be completely eliminated from the body.
B. The time it takes for half of the administered dose of the drug to be eliminated from the body.
C. The time it takes for the drug to reach its peak plasma concentration.
D. The ratio of the active drug metabolite to the parent drug.
Answer: B
Rationale: The half-life is the time required for the concentration of the drug in the body to be reduced by 50%. This is important for determining dosing intervals.

7. Which route of administration bypasses the first-pass metabolism in the liver?
A. Oral
B. Intramuscular
C. Intravenous
D. Subcutaneous
Answer: C
Rationale: Intravenous administration delivers the drug directly into the bloodstream, avoiding the hepatic first-pass effect where drugs are extensively metabolized by the liver before reaching systemic circulation. Oral drugs are subject to first-pass metabolism. IM and SC drugs are absorbed into systemic circulation but may still eventually pass through the liver.

8. A drug interaction where one drug increases the effect of another drug is known as:
A. Antagonism
B. Synergism
C. Potentiation
D. Additive effect
Answer: B
Rationale: Synergism is when the combined effect of two drugs is greater than the sum of their individual effects. Potentiation is when one drug enhances the effect of another. An additive effect is when the combined effect equals the sum of individual effects. Antagonism is when one drug reduces the effect of another.

9. The "peak" of a drug level refers to:
A. The minimum concentration of drug required to produce a therapeutic effect.
B. The maximum concentration of drug in the bloodstream after administration.
C. The time at which the drug is most concentrated in the tissues.
D. The concentration of drug in the urine at steady state.
Answer: B
Rationale: The peak drug level is the highest concentration of a drug in the blood plasma after a dose has been administered. This is important for assessing potential toxicity.

10. Which of the following is a characteristic of a Schedule I controlled substance?
A. High potential for abuse and no currently accepted medical use.
B. High potential for abuse, with accepted medical uses, but with severe restrictions.
C. Lower potential for abuse than Schedule II drugs and accepted medical uses.
D. Very low potential for abuse and widely accepted medical uses.

Answer: A

Rationale: Schedule I drugs, according to the Controlled Substances Act, have a high potential for abuse and no currently accepted medical use in treatment in the United States. Examples include heroin and LSD.

11. What is the primary purpose of obtaining a drug trough level?
    A. To assess the maximum drug concentration and potential for toxicity.
    B. To ensure the drug is reaching effective therapeutic levels.
    C. To determine the rate of drug excretion.
    D. To measure the lowest drug concentration in the bloodstream, typically just before the next dose.
    Answer: D
    Rationale: A trough level is the lowest concentration of a drug in the blood plasma, measured just before the next scheduled dose. This helps ensure that the drug concentration remains within the therapeutic range and does not fall too low between doses.

12. A patient receiving a beta-blocker for hypertension would likely experience which of the following effects?
    A. Increased heart rate
    B. Decreased blood pressure
    C. Bronchodilation
    D. Vasoconstriction
    Answer: B
    Rationale: Beta-blockers work by blocking the effects of adrenaline on the heart and blood vessels, leading to a slower heart rate and lower blood pressure. They can cause bronchoconstriction, not dilation.

13. Which herbal supplement is known to interact with warfarin (Coumadin) and potentially increase bleeding risk?
    A. St. John's Wort
    B. Ginkgo Biloba
    C. Echinacea
    D. Ginseng
    Answer: B
    Rationale: Ginkgo Biloba has antiplatelet properties that can increase the risk of bleeding, especially when taken concurrently with anticoagulants like warfarin. St. John's Wort interacts with many medications by inducing liver enzymes.

14. What is the therapeutic effect of an antiemetic?
    A. To relieve pain
    B. To prevent or relieve nausea and vomiting
    C. To reduce inflammation
    D. To lower fever
    Answer: B
    Rationale: The prefix "anti-" means against, and "emetic" relates to vomiting. Therefore, an antiemetic is a drug used to prevent or treat nausea and vomiting.

15. A drug with a narrow therapeutic index requires:
    A. Infrequent monitoring of blood levels.
    B. A wide range between the effective and toxic dose.
    C. Careful monitoring of blood levels to avoid toxicity.
    D. Administration less frequently due to a long duration of action.
    Answer: C
    Rationale: A narrow therapeutic index means that the difference between the dose that produces a therapeutic effect and the dose that causes toxicity is small. Therefore, close monitoring of drug levels is crucial to ensure safety and efficacy.

16. Which of the following is not a common side effect of opioid analgesics?
    A. Constipation
    B. Respiratory depression
    C. Diarrhea
    D. Nausea and vomiting
    Answer: C
    Rationale: Opioids are well-known for causing constipation due to their effect on the gastrointestinal tract. Diarrhea is an uncommon side effect of opioids; they typically slow down bowel motility.

17. When administering a medication via the sublingual route, the nurse should instruct the patient to:
    A. Swallow the medication immediately after placing it under the tongue.
    B. Chew the medication thoroughly before swallowing.
    C. Allow the medication to dissolve completely under the tongue without swallowing.
    D. Crush the medication and mix it with a small amount of water.
    Answer: C
    Rationale: Sublingual medications are absorbed directly into the bloodstream through the mucous membranes of the mouth. Swallowing defeats the purpose of this route and subjects the drug to first-pass metabolism.

18. What is the primary mechanism of action for most antibiotics?
    A. Blocking pain receptors
    B. Inhibiting the growth or killing of microorganisms
    C. Reducing inflammation by suppressing the immune system
    D. Increasing the heart rate
    Answer: B
    Rationale: Antibiotics are designed to target and eliminate or inhibit the growth of bacteria, thus treating bacterial infections.

19. A patient develops a rash, itching, and difficulty breathing after receiving a new medication. This is most suggestive of:
    A. A therapeutic effect
    B. An adverse effect (allergic reaction)
    C. A paradoxical reaction
    D. A placebo effect
    Answer: B
    Rationale: These symptoms are classic signs of an allergic reaction, which is a type of adverse drug effect. A paradoxical reaction is the opposite of the expected effect.

20. What is the main difference between a generic drug and a brand-name drug?
    A. Generic drugs have a different active ingredient.
    B. Brand-name drugs are always more effective.
    C. Generic drugs use a different chemical formula.
    D. Generic drugs contain the same active ingredient and are bioequivalent to the brand-name drug.
    Answer: D
    Rationale: Generic drugs are required to have the same active ingredient, dosage form, strength, route of administration, and bioequivalence as the brand-name drug. They are typically less expensive because they do not have the same research and development costs and marketing expenses.